

Progress Update – Adversarial Attacks

Lara Braukmann (457008) Alina Shcherbakova (463616)

1. What are Adversarial Examples?

- Small, often barely noticeable changes to input data that lead models to make wrong predictions.
- First identified by Szegedy et al. in the field of image classification.
- Classic examples include slightly modified images that are misclassified by neural networks.
- Interestingly, the same adversarial inputs can often fool several different models, suggesting deep, shared vulnerabilities.
- Note: Improving model behavior in such cases belongs to the field of adversarial defenses (not the main focus here), e.g.:
 - Adversarial training as a form of regularization
 - Using different model families like RBF networks

2. What are Adversarial Attacks?

- The intentional creation of inputs that mislead machine learning models.
- These attacks are typically divided into:
 - **White-box attacks:** Full access to model parameters and gradients.
 - **Grey-box attacks:** Partial access e. g. to model architecture or substitute models.
 - **Black-box attacks:** Only access to input-output behavior, no knowledge of internals.
- Attack goals:
 - **Targeted Attack:** Push input to a specific incorrect label.
 - **Untargeted Attack:** Just cause any misclassification.
- Use cases:
 - To evaluate robustness and uncover model weaknesses
 - For malicious purposes (e.g., in security-critical applications)

3. Overview of Named Attacks

- **FGSM (Fast Gradient Sign Method)** – (Proposed Goodfellow)
 - baseline for other methods
- **PGD (Projected Gradient Descent)**
 - gold standard for strong white-box attacks
- **DeepFool**
 - Original: *DeepFool: A Simple and Accurate Method to Fool Deep Neural Networks*
 - Recent extensions:
 - * Enhanced Targeted DeepFool (2025): focuses on targeted class manipulation
 - * SuperDeepFool (2025): fast and accurate minimal perturbation attack
- **One-Pixel Attack**
 - See: One Pixel Attack for Fooling Deep Neural Networks
- **L0, L1, and L ∞ -norm attacks**
 - Considered stronger
 - See: *Towards Evaluating the Robustness of Neural Networks* (Carlini & Wagner)
- **Attacks in the context of Network Security**
 - (Still deciding whether to include this section in the report)
 - In areas like Network Intrusion Detection Systems (NIDS), such attacks can pose serious risks.
 - These systems increasingly use deep learning models — and are hence vulnerable as well.
 - Attacks here differ from those in computer vision.
 - See: *Adversarial Machine Learning for Network Intrusion Detection Systems: A Comprehensive Survey*

4. Preliminary Key References

- *Explaining and Harnessing Adversarial Examples* – Goodfellow et al., 2014
- *Towards Deep Learning Models Resistant to Adversarial Attacks* – Madry et al., 2017

- *DeepFool: A Simple and Accurate Method to Fool Deep Neural Networks* – Moosavi-Dezfooli et al., 2016
- *Towards Evaluating the Robustness of Neural Networks* – Carlini & Wagner, 2017
- *Adversarial Attacks on Neural Network Policies* – Huang et al., 2017
- *How Deep Learning Sees the World: A Survey on Adversarial Attacks and Defenses* – Anonymous, 2024
- *Explanations Can Be Manipulated and Geometry Is to Blame* – Ghorbani et al., 2019
- *Visual Adversarial Examples Jailbreak Aligned Large Language Models* – Xiangyu et al., 2024

5. Next Steps

- Go through each referenced paper and check whether it is still considered state-of-the-art or if newer work has disproved or extended it.
- Organize findings into the required format and make sure all citations are consistent and properly formatted.
- Writing